AgriSense v2 - Tech Stack & System Workflow

Real-Time Environmental & Crop Health Monitoring System

1. UPGRADED TECH STACK

A zero-backend, zero-cost architecture - everything runs inside a normal web browser.

Layer / Role	Technology	What it does in the system
UI structure	HTML5	Dashboard layout: parameter cards, health gauge, 7-day forecast strip, advisory & risk panels, charts.
Styling	CSS3 (no framework)	Green farm theme, responsive grid, coloured status chips, pulsing LIVE indicator, print stylesheet.
Application logic	JavaScript ES6 (vanilla)	The brain: API fetching, health model, alert engine, irrigation & risk formulas, growth tracker, leaf analysis - one commented file (app.js).
Charts	Chart.js 4 (CDN)	Live trend charts - temperature & humidity, soil moisture & rainfall (past 24 h solid, forecast dashed).
Live data "virtual sensors"	Open-Meteo API	Free & keyless: temp, humidity, soil moisture/temp, rain, wind, ET0, dew point, VPD, solar radiation + 7-day forecast.
Image analysis *new	Canvas API	Reads leaf-photo pixels and classifies green / yellow / brown tissue percentages (leaf health analyzer).
Embedded database *new	localStorage API	Stores up to 2000 reading snapshots, alert history, sowing dates and user preferences - no DB server needed.
Farm positioning *new	Geolocation API	"My Location" button auto-detects farm GPS coordinates.
Report export *new	Print API + print CSS	One click turns the dashboard into a clean printable report / PDF.
Runtime / backend	Browser + server.py	Runs via index.html or tiny python server. No backend / DB server - by design; every layer can be swapped for real hardware later without changing the UI.

What is new in v2 vs v1

v1 : HTML + CSS + JS + Chart.js + Open-Meteo (5 technologies)
v2 adds: Canvas API | localStorage | Geolocation | Print API | new API data groups (ET0, VPD, solar).

One-line summary for presentations / viva

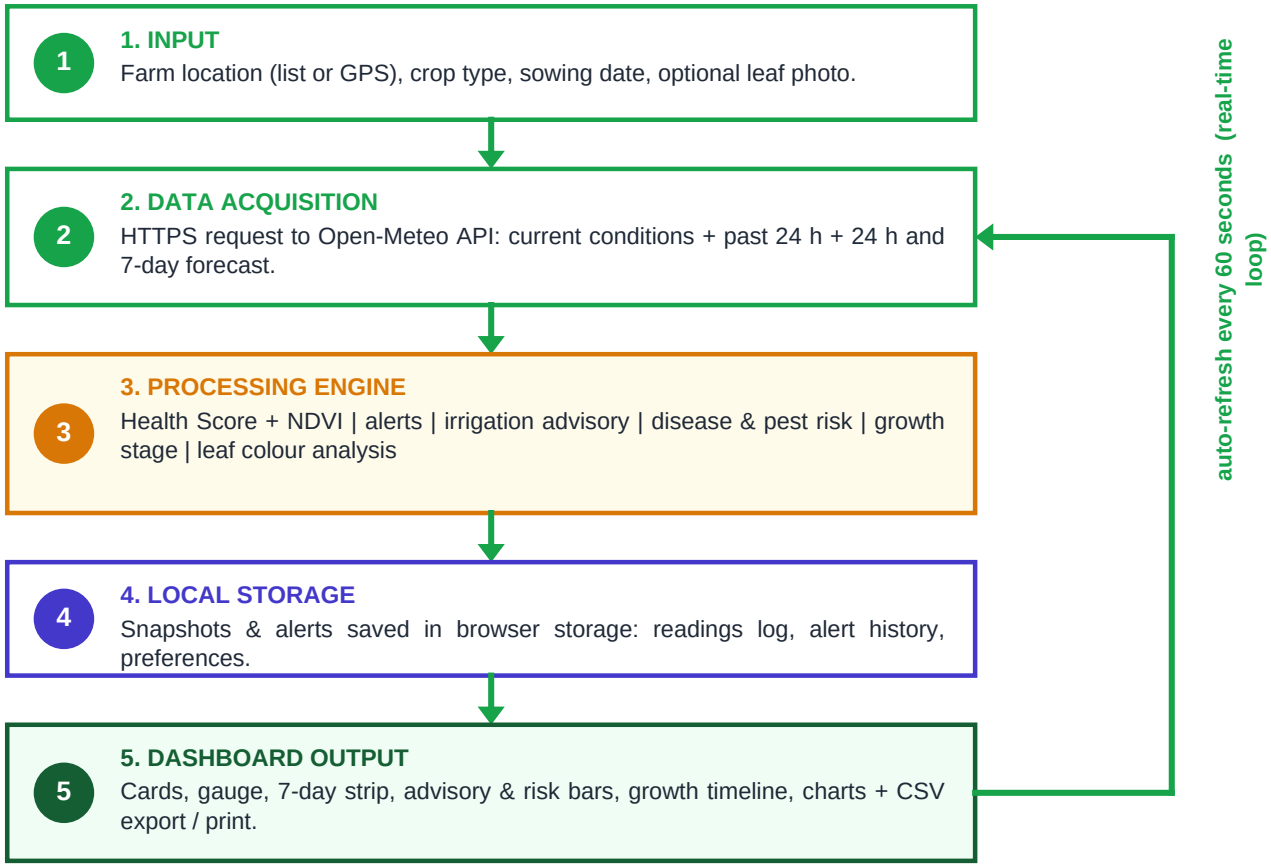
"A zero-backend architecture: Open-Meteo is the virtual sensor layer, vanilla JavaScript is the processing engine with seven software models, Canvas API handles image analysis and localStorage acts as an embedded database - zero cost, and every layer can later be swapped for real hardware or a backend without changing the UI."

AgriSense v2 - Tech Stack & System Workflow

Real-Time Environmental & Crop Health Monitoring System

2. SYSTEM WORKFLOW

The system runs as a continuous five-stage cycle, repeated automatically every 60 seconds (real-time):



Workflow in brief

1. **Input**
   Location (regions list or GPS), crop (6 options), optional sowing date & leaf photo. Selections are remembered.
2. **Data acquisition (virtual sensor layer)**
   Every 60 s the coordinates go to the Open-Meteo API, which returns current + history + forecast JSON - the same variables physical sensors measure.
3. **Processing engine (7 software models)**
   Health score & NDVI, alert rules, irrigation advisory, disease/pest risk, growth stage, leaf analysis - all in the browser.
4. **Local storage**
   Each cycle appends a snapshot and any new alert to localStorage (max 2000 readings / 60 alerts) - an embedded mini database.
5. **Dashboard output & loop**
   Results render instantly as cards, gauge, bars, timeline and charts. Log exports as CSV; one click prints a report. Repeats every 60 s.

AgriSense v2 - Tech Stack & System Workflow

Real-Time Environmental & Crop Health Monitoring System

3. MODELS, RULES & FORMULAS

3.1 Crop Health Score bands

Score = 100 - penalties for each parameter outside the crop's optimal range (clamped 5-100).

Score range	Status	Meaning
80 - 100	Excellent	All parameters near optimal - crop is healthy.
60 - 79	Good	Mild stress - monitor closely.
40 - 59	Moderate stress	Corrective action needed soon.
5 - 39	Severe stress	Immediate action required.

3.2 Smart irrigation advisory

Net irrigation (mm) = max(0, ET0 - rain expected next 24 h) + soil deficit
soil deficit = (optimal minimum moisture % - current %) x 1.2 mm per %
1 mm over 1 acre = approx. 4047 litres => shown as mm + litres/acre + advice level.

3.3 Disease & pest risk index

Risk % = (forecast hours in conducive band / 24) x 100
Fungal band: humidity >= 85% and temp >= 18 deg C | Pest band: 25-33 deg C and humidity 60-85%
Levels: below 25% LOW | 25-49% MODERATE | 50%+ HIGH

3.4 Alert rule engine (main rules)

- Soil moisture below crop optimum -> irrigation recommended
- Soil moisture far above optimum -> waterlogging risk, pause irrigation
- Temp above optimal max +3 deg C -> heat stress, irrigate / shade
- Temp below optimal min -3 deg C -> cold stress, frost protection
- Humidity > 85% and warm -> fungal disease risk, scout field
- Rain >= 2 mm expected (24 h) -> delay irrigation and spraying
- Wind > 25 km/h -> avoid pesticide spraying (drift risk)

3.5 Other advanced modules

Module	How it works
Growth stage tracker	**DAS = today - sowing date; stage from per-crop day ranges (wheat: germination 0-10, tillering 10-45, jointing 45-70, flowering 70-95, filling 95-120, maturity 120-150).**
Leaf analyzer (experimental)	**Each pixel of the photo classified green / yellow / brown; percentages give a stress verdict. ML detection = future scope.**
Data logging + CSV	**Every refresh stores a snapshot (max 2000); one click downloads a CSV for Excel and report evidence.**
Print report	**window.print() + print CSS turns the dashboard itself into a clean report page.**

Result: one page shows what sensors would measure, crop health, water need, disease/pest risk and growth stage - updated live, logged automatically, exportable as CSV or printout. Zero hardware cost.